

CLAIMS

1. ~~A therapeutic agent method of treating a cancer patient comprising administering to the patient a therapy~~ that inhibits poly [ADP-ribose] polymerase (“anti-PARP therapy”) for use in a method of treating a cancer patient, wherein the anti-PARP therapy is administered independent of DNA repair status, optionally independent of BRCA status, or

2. ~~A method of treating a cancer patient comprising administering to the patient a therapy that inhibits poly [ADP-ribose] polymerase (“anti-PARP therapy”)~~, wherein the anti-PARP therapy is commenced prior to determining the DNA repair status of the patient or BRCA status of the patient, or

3. ~~A method of treating a cancer patient comprising administering to the patient a therapy that inhibits poly [ADP-ribose] polymerase (“anti-PARP therapy”)~~, wherein the anti-PARP therapy is commenced in the absence of determining the DNA repair status of the patient or BRCA status of the patient.

24. The therapeutic agent for use according to method of any one of claims 1-3, wherein the patient is characterized by an absence of a mutation in BRCA1 and/or BRCA2;

5. ~~The method of any one of claims 1-3, wherein the patient has at least one mutation in BRCA1 and/or BRCA2.~~

63. A ~~therapeutic agent method of treating a cancer patient comprising administering to the patient a therapy~~ that inhibits poly [ADP-ribose] polymerase (“anti-PARP therapy”) for use in a method of treating a cancer patient, wherein the patient is characterized by an absence of a mutation in BRCA1 and/or BRCA2, and wherein the cancer is a gynecological cancer selected from the group consisting of: ovarian cancer, fallopian tube cancer, peritoneal cancer, and breast cancer, or

~~7. — A method of treating a cancer patient comprising administering to the patient a therapy that inhibits poly [ADP-ribose] polymerase (“anti-PARP therapy”), wherein the cancer is characterized by an absence of a mutation in BRCA1 or BRCA2, and wherein the cancer is a gynecological cancer selected from the group consisting of: ovarian cancer, fallopian tube cancer, peritoneal cancer, and breast cancer.~~

~~48. The therapeutic agent for use according to method of any one of claims 1-37, wherein the anti-PARP therapy is administered at a dose equivalent to about 100 mg, about 200 mg, or about 300 mg of niraparib or a salt or derivative thereof.~~

~~59. The therapeutic agent for use according to method of any one of claims 1-48, wherein the anti-PARP therapy comprises administration of an agent that inhibits PARP-1 and/or PARP-2.~~

~~10. — The method of claim 9, wherein the agent is a small molecule, a nucleic acid, a polypeptide (e.g., an antibody), a carbohydrate, a lipid, a metal, or a toxin.~~

~~11. — The method of claim 10, wherein the agent is selected from the group consisting of: ABT-767, AZD-2461, BGB-290, BGP-15, CEP-9722, E7016, E7449, fluzoparib, INO1001, JPI-289, MP-124, niraparib, olaparib, ONO2231, rucaparib, SC-101914, talazoparib, veliparib, WW-46, and salts or derivatives thereof.~~

~~12. — The method of claim 11, wherein the agent is selected from the group consisting of: niraparib, olaparib, rucaparib, talazoparib, veliparib, and salts or derivatives thereof.~~

~~136. The therapeutic agent for use according to method of claim 15, wherein the agent is niraparib or a salt or derivative thereof.~~

~~147. The therapeutic agent for use according to method of any one of claims 1-13 or claim 2, wherein the cancer is a gynecological cancer, optionally~~

~~15.~~ The method of claim 14, wherein the gynecological cancer is selected from the group consisting of: ovarian cancer, fallopian tube cancer, peritoneal cancer, and breast cancer.

~~168.~~ The therapeutic agent for use according to method of any one of claims 1-~~7~~15, wherein the cancer is a recurrent cancer.

~~917.~~ The therapeutic agent for use according to method of any one of claims 1-~~168~~, wherein the method prolongs progression free survival as compared to control, and/or

~~18.~~ The method of any one of claims 1-17, wherein the method reduces the hazard ratio for disease progression or death as compared to control, and/or

~~19.~~ The method of any one of claims 1-18, wherein the method prolongs overall survival as compared to control, and/or

~~20.~~ The method of any one of claims 1-19, wherein the method achieves an overall response rate of at least 30%.

~~21.~~ The method of any one of claims 1-16, wherein the anti-PARP therapy is administered in a regimen determined to achieve i) prolonged progression free survival as compared to control, ii) a reduced hazard ratio for disease progression or death as compared to control, iii) prolonged overall survival as compared to control, or iv) an overall response rate of at least 30%.

1022. A therapeutic agent for use in a method of treating a patient having a recurrent cancer selected from ovarian cancer, fallopian tube cancer, or peritoneal cancer, wherein the therapeutic agent is niraparib, the method comprising administering niraparib to the patient according to a regimen determined to achieve i) prolonged progression free survival as compared to control, ii) a reduced hazard ratio for disease progression or death as compared to control, iii) prolonged overall survival as compared to control, or iv) an overall response rate of at least 30%.

~~1123.~~ The therapeutic agent for use according to method of any one of claims 1-1022, wherein the cancer is ovarian cancer (e.g. wherein the patient has high grade serous ovarian cancer or high grade predominantly serous histology ovarian cancer), optionally

~~24.~~—The method of claim 23, wherein the ovarian cancer is platinum sensitive at the commencement of treatment, or

~~25.~~—The method of any one of claims 1-22, wherein the cancer is fallopian tube cancer, optionally

~~26.~~—The method of claim 25, wherein the fallopian tube cancer is platinum sensitive at the commencement of treatment, or

~~27.~~—The method of any one of claims 1-22, wherein the cancer is peritoneal cancer, optionally

~~28.~~—The method of claim 27, wherein the peritoneal cancer is platinum sensitive at the commencement of treatment.

~~29.~~—The method of any one of claims 1-3, 5, and 8-28, wherein the patient has at least one mutation selected from (i) a germline mutation in BRCA1 and/or BRCA2, or (ii) a sporadic mutation in BRCA1 and/or BRCA2.

~~30.~~—The method of claim 29, wherein the patient has a germline mutation in BRCA1 and/or BRCA2.

~~31.~~—The method of claim 29, wherein the patient has a sporadic mutation in BRCA1 and/or BRCA2.

1232. The therapeutic agent for use according to method of any one of claims 1-~~114~~ and ~~6-28~~, wherein the patient is characterized by an absence of a mutation in BRCA1 and/or BRCA2, optionally

~~33.~~—~~The method of claim 32,~~ wherein the patient is characterized by an absence of a germline mutation in BRCA1 and/or BRCA2, or

~~34.~~—~~The method of claim 32,~~ wherein the patient is characterized by an absence of a sporadic mutation in BRCA1 and/or BRCA2.

1335. The therapeutic agent for use according to method of any one of claims 1-~~3, 5,~~ and ~~48-1134~~, wherein the patient has a tumor with a positive homologous recombination deficiency status.

1436. The therapeutic agent for use according to method of any one of claims 1-~~124, 6-28,~~ and ~~32-34~~, wherein the patient has a tumor with a negative homologous recombination deficiency status.

1537. The therapeutic agent for use according to method of any one of claims ~~1021-1436~~, wherein the regimen is determined to achieve a hazard ratio for disease progression of less than about 0.5, optionally

~~38.~~—~~The method of claim 37,~~ wherein the hazard ratio for disease progression is less than about 0.3, and/or

~~39.~~—~~The method of any one of claims 21-38,~~ wherein the regimen is determined to achieve prolonged progression free survival of at least 9 months, optionally

~~40.~~—~~The method of claim 39,~~ wherein the prolonged progression free survival is at least 12 months, especially

41. ~~The method of claim 40,~~ wherein the prolonged progression free survival is at least 21 months.

42. ~~The method of claim 23 wherein the patient has high grade serous ovarian cancer or high grade predominantly serous histology ovarian cancer.~~

1643. ~~The therapeutic agent for use according to method of any one of claims 1-1542,~~ wherein the method comprises continued treatment until disease progression or unacceptable toxicity, optionally

44. ~~The method of claim 43,~~ wherein the method comprises reducing the dose of the anti-PARP therapy or niraparib in response to toxicity, especially

45. ~~The method of claim 44,~~ wherein the method comprises reducing the dose of the anti-PARP therapy from about a dose equivalent to about 300 mg to a dose equivalent to about 200 mg, or

46. ~~The method of claim 44,~~ wherein the method comprises reducing the dose of niraparib from about 300 mg to about 200 mg.

1747. ~~The therapeutic agent for use according to method of any one of claims 1-1644,~~ wherein the method comprises administering a plurality of oral doses, optionally

48. ~~The method of claim 47,~~ wherein the method comprises once daily (QD) dosing, and/or

49. ~~The method of any one of claims 1-48,~~ wherein the method comprises at least one 28 day cycle of dosing, and/or

50. ~~The method of any one of claims 47-49,~~ wherein the oral dose is administered in one or more unit dosage forms, optionally

51. ~~The method of claim 50,~~ wherein the one or more unit dosage forms are capsules.

~~1852. The therapeutic agent for use according to method of any of claims 1-37, 59-1543, and 1747-51,~~ wherein the oral dose is an amount within a range equivalent to about 5 to about 400 mg of niraparib, optionally

~~53. The method of claim 52,~~ wherein the amount is equivalent to about 5, about 10, about 25, about 50, about 100, about 150, about 200, about 250, about 300, about 350, or about 400 mg of niraparib,

~~54. The method of claim 53,~~ especially wherein the amount is equivalent to about 300 mg of niraparib, or

~~55. The method of claim 53,~~ wherein the amount is equivalent to about 200 mg of niraparib, or

~~56. The method of claim 53,~~ wherein each unit dosage form comprises an amount equivalent to about 100 mg, about 200 mg, or about 300 mg of niraparib.

~~1957. The therapeutic agent for use according to method of claim 1748 or claim 56,~~ wherein each QD dose is equivalent to about 100 mg, about 200 mg, or about 300 mg of niraparib, or

~~58. The method of claim 48,~~ wherein each QD dose is administered as three unit dosage forms equivalent to about 100 mg of niraparib.

~~2059. The therapeutic agent for use according to method of any one of claims 917-1958,~~ wherein progression free survival is characterized by a complete response of one or more target tumors, or

~~60. The method of any one of claims 17-58,~~ wherein progression free survival is characterized by a partial response of one or more target tumors.

~~2161.~~ The therapeutic agent for use according to method of any one of claims 1-~~2060~~, wherein the method comprises administration to the patient in a fasted state~~;~~, or

~~62.~~ ~~The method of any one of claims 1-60,~~ wherein the method comprises administration to the patient in a fed state.

~~2263.~~ The therapeutic agent for use according to method of any one of claims ~~917-2162~~, wherein the prolonged progression free survival is at least 9 months~~;~~, optionally

~~64.~~ ~~The method of claim 63,~~ wherein the prolonged progression free survival is at least 12 months~~;~~,

~~65.~~ ~~The method of claim 64, wherein the prolonged progression free survival is at least 15 months;~~,

~~66.~~ ~~The method of claim 65, wherein the prolonged progression free survival is at least 18 months;~~,

~~67.~~ ~~The method of claim 66, wherein the prolonged progression free survival is at least 21 months;~~,

~~68.~~ ~~The method of claim 67, wherein the prolonged progression free survival is at least 24 months;~~,

~~69.~~ ~~The method of claim 68, wherein the prolonged progression free survival is at least 27 months;~~,

~~70.~~ ~~The method of claim 69, wherein the prolonged progression free survival is at least 30 months;~~,

~~71. The method of claim 70, wherein the prolonged progression free survival is at least 33 months, or~~

~~72. The method of claim 71, wherein the prolonged progression free survival is at least 36 months.~~

~~2373. The therapeutic agent for use according to method of any one of claims 1048-2272, wherein the hazard ratio for disease progression is about 0.3, or~~

~~74. The method of claim 18-72, wherein the hazard ratio for disease progression is about 0.45, or~~

~~75. The method of claim 18-72, wherein the hazard ratio for disease progression is about 0.5, or~~

~~76. The method of claim 18-72, wherein the hazard ratio for disease progression is less than about 0.5, optionally~~

~~77. The method of claim 76, wherein the hazard ratio for disease progression is less than about 0.45,~~

~~78. The method of claim 77, wherein the hazard ratio for disease progression is less than about 0.4,~~

~~79. The method of claim 78, wherein the hazard ratio for disease progression is less than about 0.35, or~~

~~80. The method of claim 79, wherein the hazard ratio for disease progression is less than about 0.3,~~

2484. The therapeutic agent for use according to method of any one of claims 1-2380, wherein anti-PARP therapy or niraparib is administered as a maintenance therapy.

2582. A therapeutic agent for use in a method of treating administering niraparib to a patient lacking homologous repair deficiency and having a recurrent cancer selected from platinum-sensitive ovarian cancer, fallopian tube cancer, or primary peritoneal cancer, wherein the therapeutic agent is niraparib, the method comprising administering niraparib to the patient according to a regimen determined to achieve prolonged progression free survival as compared to control.

2683. The therapeutic agent for use according to method of any one of claims 1-2265, wherein the method has further been determined to achieve improved progression free survival 2 as compared to control.

2784. The therapeutic agent for use according to method of any one of claims 1-2683, wherein the method has further been determined to achieve improved chemotherapy free interval as compared to control, and/or

85. ~~The method of any one of claims 1-84,~~ wherein the method has further been determined to achieve improved time to first subsequent therapy as compared to control, and/or

86. ~~The method of any one of claims 1-85,~~ wherein the method has further been determined to achieve improved time to second subsequent therapy as compared to control, and/or

87. ~~The method of any one of claims 1-86,~~ wherein the method has been determined to not have a detrimental effect on Quality of Life as determined by FOSI and/or EQ-5D-5L, and/or

88. ~~The method of any one of claims 1-87,~~ wherein the method has been determined to not impact the effectiveness of a subsequent treatment with a chemotherapeutic agent, optionally

~~89. The method of claim 88, wherein the next chemotherapeutic agent is a platinum agent,~~
especially

~~90. The method of claim 89, wherein the platinum agent is selected from cisplatin,~~
carboplatin, oxaliplatin, nedaplatin, triplatin tetranitrate, phenanthriplatin, picoplatin, or
satraplatin.